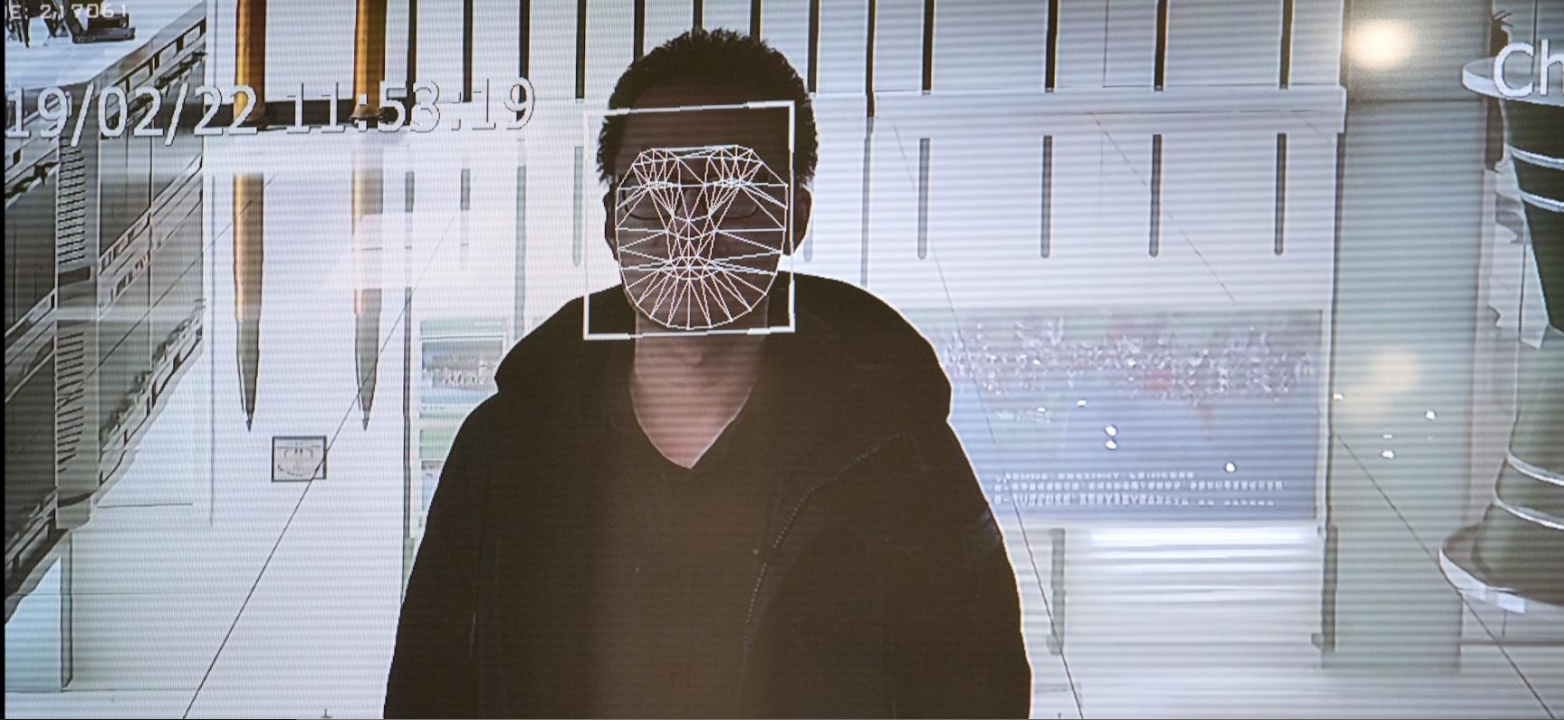
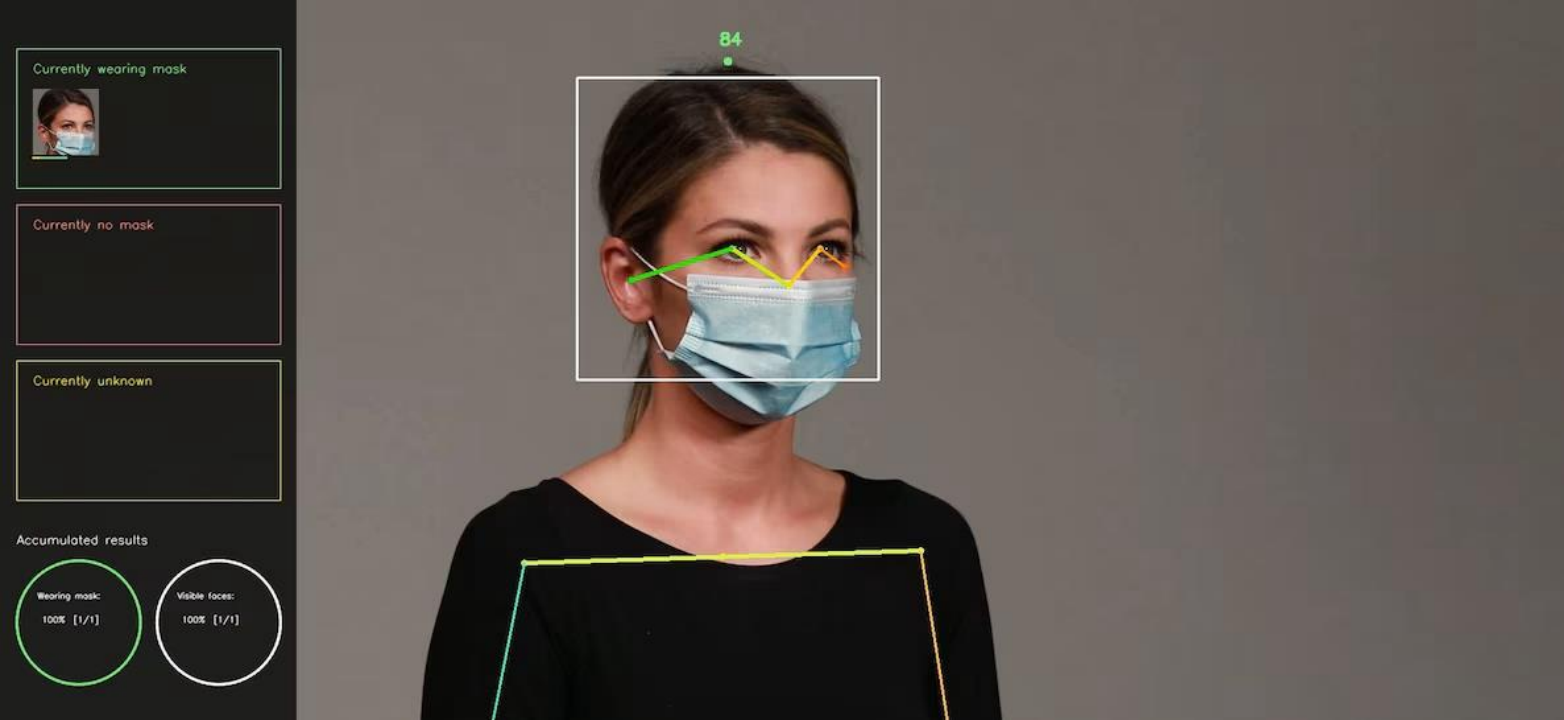




FACE RECOGNITION UNDER OCCLUSION

- RICCARDO DEIDDA
M:70/90/00639.



THE PROBLEM

MAIN ISSUES:

- ◆ Feature extraction.
- ◆ Face recognition with only partial features.



BASICS OF FACE RECOGNITION

Face recognition fundamental steps:

- ◈ Image normalization and alignment.
- ◈ Feature extraction.
- ◈ Results analysis.



POSSIBLE APPROACHES

Feature extraction for face recognition can be approached through three main categories:

- ◇ Handcrafted features
- ◇ Texture descriptors
- ◇ Convolutional Neural Networks

HANDCRAFTED FEATURES

Traditional handcrafted approaches include methods such as:

- ◆ Haar Cascade Classifier.
- ◆ Histogram of Oriented Gradients.

While computationally efficient, they have relatively basic ability in discrimination between identities, so more robust systems must be applied.

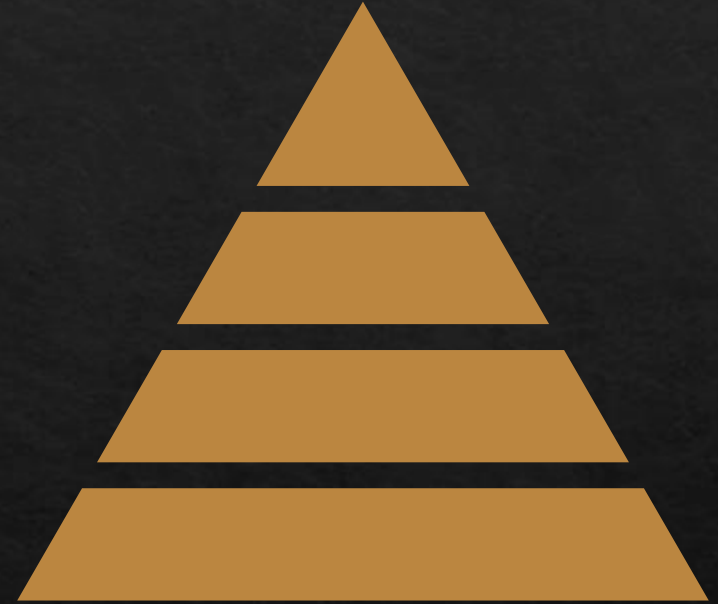


TEXTURAL DESCRIPTORS

Textural descriptors are techniques aimed at capture textural patterns in the face region. Examples of textural descriptors are:

- ◊ Eigenfaces
- ◊ Local Binary Patterns
- ◊ and Binarized Statistical Image Features

These descriptors often achieve better performance than basic handcrafted methods while retaining relatively low computational complexity



CONVOLUTIONAL NEURAL NETWORKS

Modern face recognition systems rely on deep neural networks to learn highly discriminative representations.

- ◇ Deliver **robust and accurate results** when well implemented
- ◇ Require **more computation** than simpler methods

CNNs can be implemented in two different way:

- ◇ From scratch with Tensorflow or Keras.
- ◇ With Transfer Learning trough libraries such as Deepface.



OUR DATASET



81 IDENTITIES WITH 1 SAMPLE EACH

IMAGE PREPROCESSING

Image preprocessing should include normalization and enhancement. RetinaFace is a **state-of-the-art tool** for face detection and alignment. However, it **missed some faces** during testing...

Fortunately, the **current dataset is already aligned**, so preprocessing is simplified.





1- [ID:100]
Distance: 0.2292
Score: 0.6257



2- [ID:306]
Distance: 0.3503
Score: 0.3643



3- [ID:340]
Distance: 0.3648
Score: 0.396



FIRST IMPLEMENTATION

IMPLEMENTATION STRUCTURE:

- ◇ **BSIF** used for feature extraction, with **CLAHE** preprocessing.
- ◇ Applied on an **artificially masked dataset** to compare only visible features.

RESULTS:

- ◇ Probe was correctly identified.
- ◇ Manhattan metrics yielded the best results.
- ◇ Low separation margin for Manhattan metrics.
- ◇ Algorithm was **very fast**

SECOND IMPLEMENTATION

IMPLEMENTATION STRUCTURE:

- ❖ DeepFace-based **CNN** for feature extraction
- ❖ **Facenet512** was used as the model backbone. (other models were available but gave no significant improvements)
- ❖ RetinaFace used for preprocessing.
- ❖ Tested on an **artificially masked dataset** to compare only visible features.

RESULTS:

- ❖ Probe was correctly identified.
- ❖ Good separation margin in both scores and distances.
- ❖ **Very slow performances:** up to **20 minutes** for feature extraction and processing.

1- [ID:100]
Distance: 0.0043
Score: 0.9966



2- [ID:31]
Distance: 0.5111
Score: 0.6009



3- [ID:102]
Distance: 0.5965
Score: 0.5342



FINAL IMPLEMENTATION

IMPLEMENTATION STRUCTURE:

- ◊ DeepFace-based **CNN** for feature extraction
- ◊ **Facenet512** was used as the model backbone.
- ◊ RetinaFace used for preprocessing.
- ◊ Tested on an **artificially masked dataset** to compare only visible features.

MAIN DIFFERENCE:

- ◊ Deepface only performed feature extraction, all other computation where carried out by locally defined functions.

RESULTS:

- ◊ Very high accuracy, same results as the previous deepface implementation.
- ◊ Computation time was cut in half by 10 minutes.



RESULT ANALYSIS

This implementation shows strong results across the board, even when tested with other masked reference images. However, we're comparing masked images to their unmasked counterparts, which can inflate performance.



With too few samples to form a proper dataset, this setup likely encourages overfitting and may perform poorly in real-world scenarios.

SUGGESTED SOLUTIONS

What can be done to improve our situation?

- ◇ Increase the **number of samples** per class
- ◇ **Use real masked faces** or apply **different synthetic masks** to gallery vs. probe
- ◇ **Store multiple views per identity** and use **template/centroid averaging**
- ◇ **Improve visible-region recognition** with more discriminative algorithms and not just cropping.
- ◇ Implement a **robust face alignment step** before visible-region recognition or **enforce dataset quality standards** rejecting unusable or nonaligned images.

-RICCARDO DEIDDA

THANK YOU FOR
LISTENING!